

Identifying the Disability Belt and Epicenters in the American Nations

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Human health in the United States faces several ongoing challenges, including chronic disease and unhealthy living behavior pandemics, especially physical inactivity (PI). We have previously demonstrated heterogeneity in PI-chronic disease prevalence across the United States and the existence of regional *hot spots* and PI-chronic disease synergy concentrations.¹ The prevalence and distribution patterns of individuals with disability in the United States, a factor related to the unhealthy living-chronic disease continuum, require further exploration.

We obtained county-level, 2021 age-adjusted disability (ie, any disability among adults aged ≥ 18 yr) and no leisure time physical activity (ie, PI) prevalence from the 2023 Centers of Disease Control and Prevention PLACES database.² Both disability and PI were obtained through the Behavioral Risk Factor Surveillance System survey.³ To determine any disability, respondents were asked whether they had disability related to hearing, vision, cognition, mobility, self-care, or independent living using a standardized six-item questionnaire.⁴ Physical inactivity was defined as answering “no” to the following question: “During the past month, other than your regular job, did you participate in any physical activities or exercises such as running, calisthenics, golf, gardening, or walking for exercise?”⁵ The American Nations regional data set was obtained from the Nationhood Lab.⁶ Briefly, the American Nations model, which has been detailed previously,⁶ discerns and maps North American regional cultures based on “first settler effects” and competing colonization patterns, which do not respect state or even international boundaries.⁷ The PLACES and Nationhood Lab databases were linked through zip code identifiers.

The Infographic illustrates for the 2021 age-adjusted prevalence of disability in the American Nations, both as an overall prevalence in each region and at the county level, clear heterogeneity in disability prevalence exists. Within the continental United States, there is an apparent *disability belt* expanding across the southern regions that encompass several of the American Nations, including Greater Appalachia, Deep South, El Norte, and New France. Epicenters of a particularly high prevalence within the belt exist within the Deep South and Greater Appalachia regions. The First Nation region in Alaska also demonstrates a disproportionately high prevalence of disability. The pattern of disability prevalence illustrated in the Infographic is strikingly similar to patterns of PI that we have previously reported.⁷ The Pearson bivariate correlation revealed a highly significant ($r = 0.87$, $P < .001$) relationship between the 2021 county-level prevalence of age-adjusted PI and disability.

Linear regression analysis produced the following equation, highly predictive of age-adjusted PI on a county level: County-level age-adjusted PI (%) = $5.763 + \text{Age-adjusted county-level disability prevalence (\%)} \times 1.976$ ($R^2 = 0.72$, standard error of estimate: 2.8%, $P < .001$).

The observations herein further reinforce regional health disparities impacting the United States. The visualization of disability prevalence is consistent with previous findings related to unhealthy living behaviors and characteristics.¹ We were not surprised to find a significant county-level relationship between PI and disability prevalence, but the strength of this relationship was unexpected; clearly, PI and disability prevalence combine to create a disconcerting elevated risk for poor health outcomes and low quality of life. The particular crisis of PI in people with disabilities has been recognized for some time; unfortunately, there has been insufficient progress made in addressing this issue.⁸ Furthermore, our findings indicate unique cultural factors, according to the American Nations model,⁷ play a role in both PI¹ and disability prevalence. A better understanding of unique driving forces for health challenges at a regional/local level is needed.

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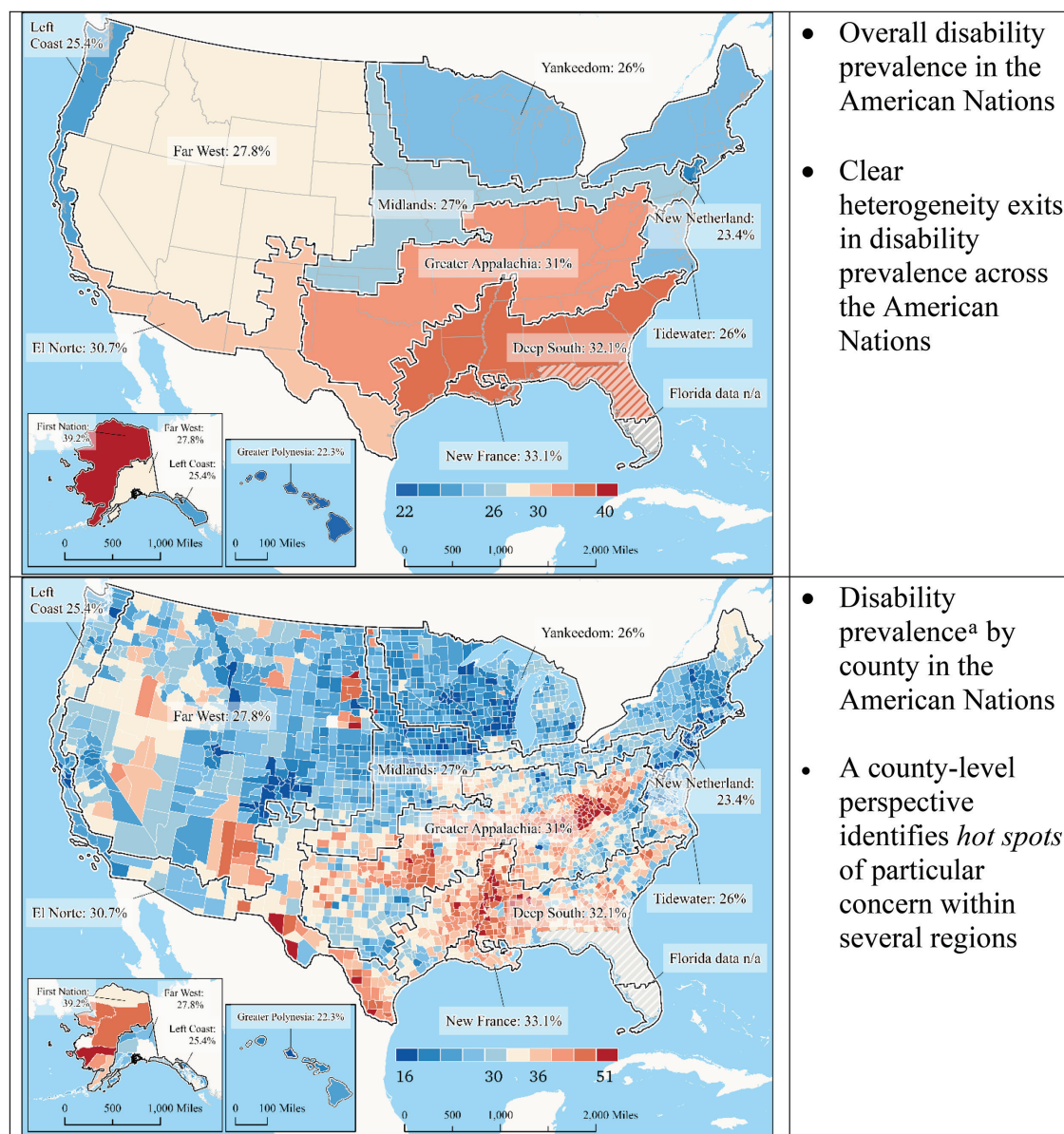
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- Overall disability prevalence in the American Nations
- Clear heterogeneity exists in disability prevalence across the American Nations
- Disability prevalence^a by county in the American Nations
- A county-level perspective identifies *hot spots* of particular concern within several regions

Figure. Prevalence of any age-adjusted disability among adults aged ≥18 yr in the American Nations. Abbreviation: BRFSS, Behavioral Risk Factor Surveillance System. ^aEstimates for Florida are not available for measures based on BRFSS 2021. Florida was unable to collect data over enough months to meet the minimum requirements for inclusion in the 2021 annual aggregate data set (<https://www.cdc.gov/places/help/data-notes>). This figure is available in color online (www.jcrjournal.com).